Daily Growth at Risk: financial or real drivers? The answer is not always the same

Helena Chuliá, Ignacio Garrón and Jorge M. Uribe

Riskcenter-IREA & Universitat de Barcelona

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Motivation I

- Growth at Risk (GaR) relates future GDP downside risks to financial conditions.

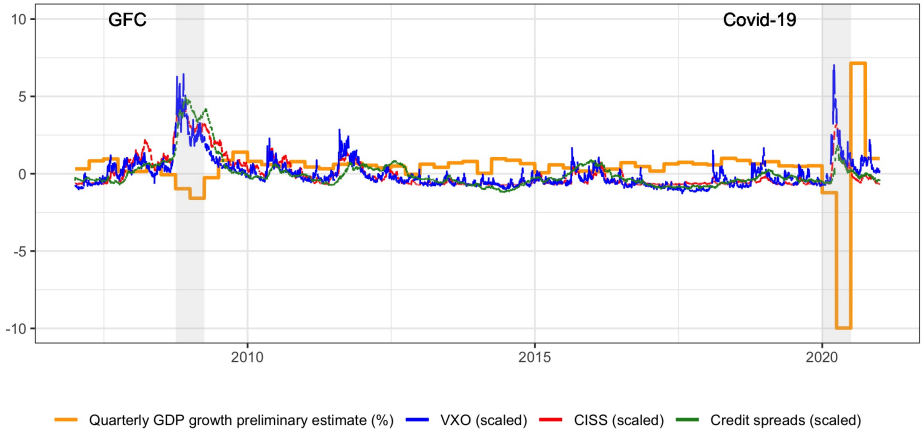

Figure 1: GDP growth vs High-frequency financial indicators

Motivation I

- GaR has been applied globally.

  Arrigoni et al. 2020, Figueres and Jarociński (2020), Brownlees et al (2021), Prasad et al., (2019).

- To gauge inflation and unemployment risks.

  Adams et al. (2022), Kiley (2022), López-Salcedo and Loria (2022).

Motivation I

- Are financial conditions the only source of GDP downside risks? What about real variables?

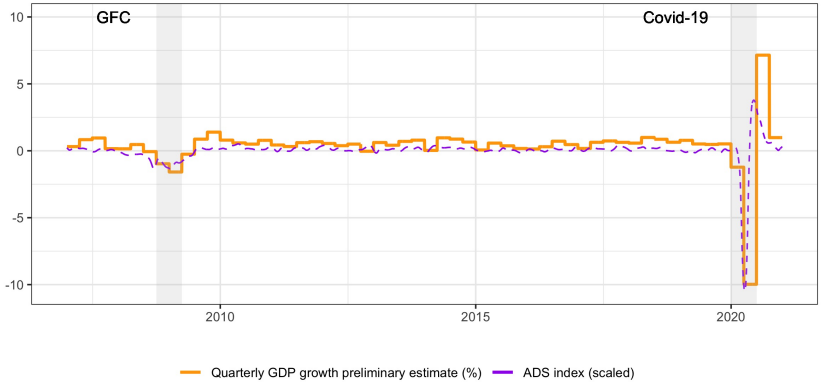

Figure 2: GDP growth vs High-frequency real indicator

Motivation I

- Are financial conditions the only source of GDP downside risks? What about real variables?
- After controlling for real variables, financial indicators have little to add to the mix.

  Reichlin et al., (2020), Plagborg-Møller et al., (2020).

Motivation II

- Financial conditions are usually taken as quarterly averages, but have higher frequency!

  Adrian et al. (2019), Brownlees et al (2021), Adams et al. (2022)

- Should we use instead high-frequency financial indicators (i.e, daily or weakly)?

  Carriero et al. (2022), Ferrara et al. (2021).

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This paper

Real-time daily GaR nowcasts using both daily financial and real variables are shown to be important to gauge GDP risks.

- High-frequency indicators (12): 11 financial indicators and 1 real indicator.
- Seven different models.
- Forecast combination is applied to get a combined-GaR estimate.
- Evaluation 1: traditional GaR vs our framework.
- Evaluation 2: Individual GaR vs combined-GaR.

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Data

Pseudo-real-time sample spans the period from Jan. 1, 1986 to Dec. 31, 2020.

1. Real Gross Domestic Product (GDP) collected in real-time.
2. ADS index vintages collected in real-time from November 30, 2008.
3. Interest rate spread (ISPREAD).
4. Effective Federal Funds Rate (EEFR).
5. Credit spread (CSPREAD).
6. Excess return on the market (RET).
7. Returns on the portfolio of small minus big stocks (SMB).
8. Returns on the portfolio of high minus low book-to-market ratio stocks (HML).
9. Returns on a winner minus loser momentum spread portfolio (MOM).
10. CBOE SP 100 Volatility Index (VXO).
11. Term spread (TERM).
12. Spread between 3-Month LIBOR based on US dollars and 3-Month Treasury Bill (TED).
13. Composite Indicator of Systemic Stress (CISS).

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Methodology

- We extend Growth at Risk framework (Adrian et al., 2019) to account daily flow of information up to T.

$$\underbrace{Q_{\tau=0.10}(y_T)}_{\text{GDP growth}} = \beta_0(\tau) + \underbrace{\beta_1(\tau)y_{T-1}}_{\text{lagged GDP growth}} + \underbrace{X'_{T-h_d}\delta(\tau)}_{\text{Daily indicator}}$$

- In general, we want to produce:

$$Q_{0.10}(y_T|y_{T-1}, X_{T-h_d})$$

- where $Q_{0.10}(.)$ comes from a mixed frequency model.

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Methodology

- Let X_{t-h_d} be a vector of 1 year daily lags of the high-frequency indicator.

$$X_{t-h_d} = [x_t^1, x_{t-1/60}^2, x_{t-2/60}^3, x_{t-3/60}^4, \ldots, x_{t-239/60}^{240}]'$$

- Parameter proliferation problem (240 parameters)!

Methodology

- In this setting, recall that the objective function to get the parameters is the minimization of the tick loss (TL) function.

$$TL_\tau = \frac{1}{T}\sum_{t}^{T}[\rho_\tau(y_t - \widehat{Q_\tau}(y_t))]$$

where $\rho_\tau = (1-\tau)\mathbb{1}_{e_t<0}|e_t| + \tau\mathbb{1}_{e_t>0}|e_t|$.

Models for $Q_{0.10}(.)$

1. Mixed data sampling quantile model (MIDAS-Q) with and Almon lag Polinomial (Mogliani and Simoni, 2021).
2. Bayesian MIDAS (BMIDAS-Q), similar to Ferrara et al. (2021)
3. LASSO quantile (Belloni and Chernozhukov, 2011).
4. Elastic Net quantile (Zou and Hastie, 2005).
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LASSO (best model)

- The objective function is:

$$min_{\phi} E[TL_{\tau}(\phi) + \underbrace{\alpha\lambda\frac{\sqrt{\tau(\tau}}{T}\Sigma_{j=1}^{P}|\phi(\tau)_j|]}_{LASSO}$$

- The optimal level of λ_{τ} (LASSO penalization) is calculated as in Belloni and Chernozhukov (2011).
- Higher λ means higher penalization.

LASSO selection of X_{t-hd}

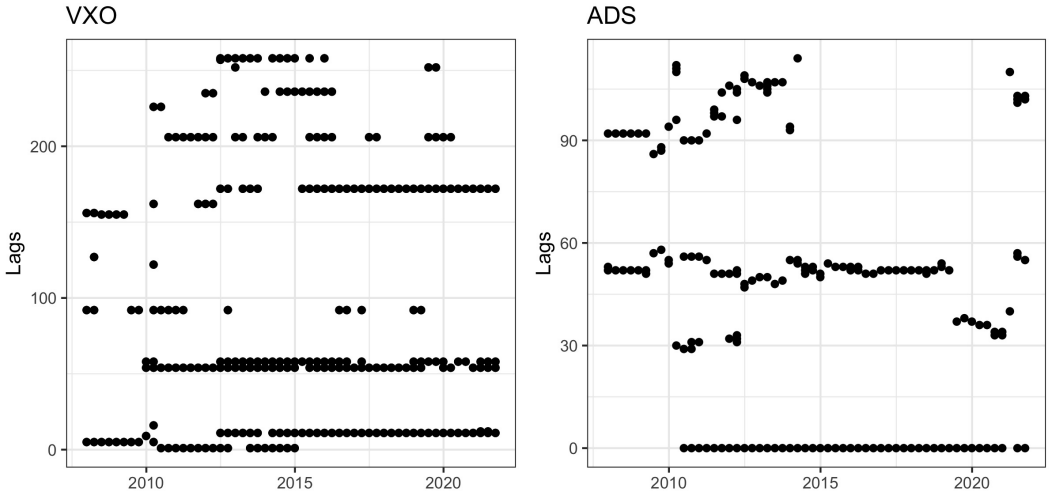

Figure 3: Lasso selection by the end of quarter

Adaptive sparse group LASSO (ASGL)-Mendez-Civieta et al. (2021)

- The objective function is:

$$min_{\phi} E[TL_{\tau}(\phi) + \underbrace{\alpha\lambda\Sigma_{j=1}^{K} w_j|\phi(\tau)_j|}_{LASSO*} + \underbrace{(1-\alpha)\lambda\Sigma_{i=1}^{M}\sqrt{p_l}v_l||\phi(\tau)^l||_2}_{\text{sparse group LASSO}}]$$

- $\tau = 0.10$.
- w_j is the weight for the j-th parameter.
- v_l is the weight for the l-th group of parameters (or high-frequency variable).
- $\alpha =$ LASSO vs sparse group LASSO.

Forecast combination

We compute the combined GaR recursively for each model (except for the ASGL-Q) as follows:

$$combined - GaR_T^* = \Sigma_i w_{i,T-h_d} * Q_{0.10}(y_T, Model_i)$$

$w_{i,T-h_d}$ Discounted TL function as in (Stock and Watson, 2004; Andreou et al, 2013), but using the tick loss as the objective function.

Algorithm for estimating combined-GaR (except ASGL)

— For each model out of 6 models do:

—— For each high-frequency indicator out of 12 do:

——— For each day starting from January 1, 2007, do:

———— Step 1: Estimate $Q^T_{0.10}(y_T|y_{T-1}, X_i)$ and produce nowcast.

———— Step 2: Calculate combination weights $Q^T_{0.10}(y_T|y_{T-1}, X_i)$.

—— Step 3: Compute individual-GaR.

— Step 4: Use optimal weights to compute combined-GaR.

Algorithm for estimating combined-GaR with ASGL

——— For each day starting from January 1, 2007, do:

——— Step 1: Calculate group LASSO weights based on regression on a subset of principal components (Mendez-Civieta et al., 2021).

——— Step 2: Estimate combined-GaR directly via $Q^T_{0.10}(y_T|y_{T-1}, X^{ASGL})$ and produce nowcast.

Forecast evaluation

1. TL function (primary criteria).
2. Unconditional coverage test.
3. Dynamic quantile test (Engle and Manganelli, 2004).

Empirical Results

- Best model: LASSO-Q.
- Benchmark (as traditional GaR): individual GaR that only include the CISS indicator.
- Benchmark model (ASGL): does not use forecast combination approach.

Nowcasting Daily GaR (starting from January 1, 2007)

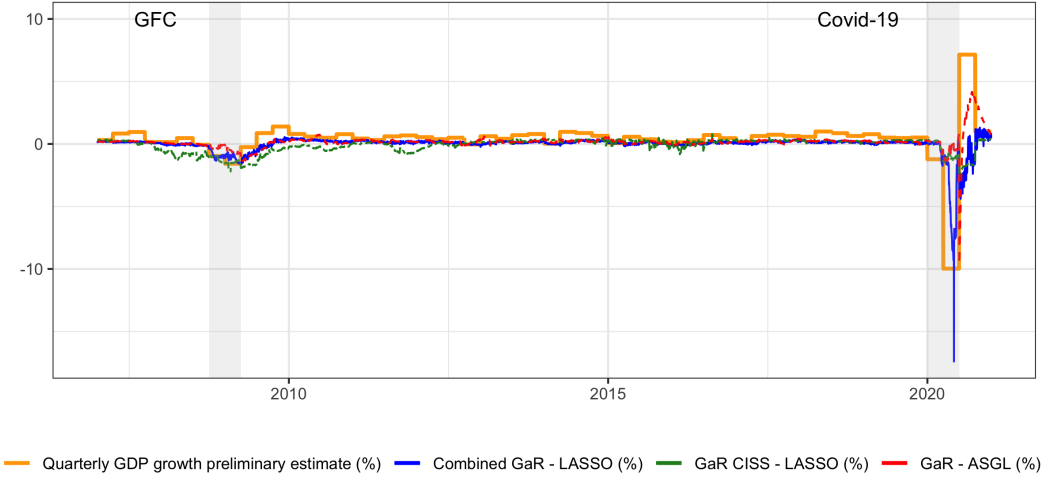

Figure 4: GaR results for LASSO-Q and AGLS-Q

Daily combination weights for LASSO

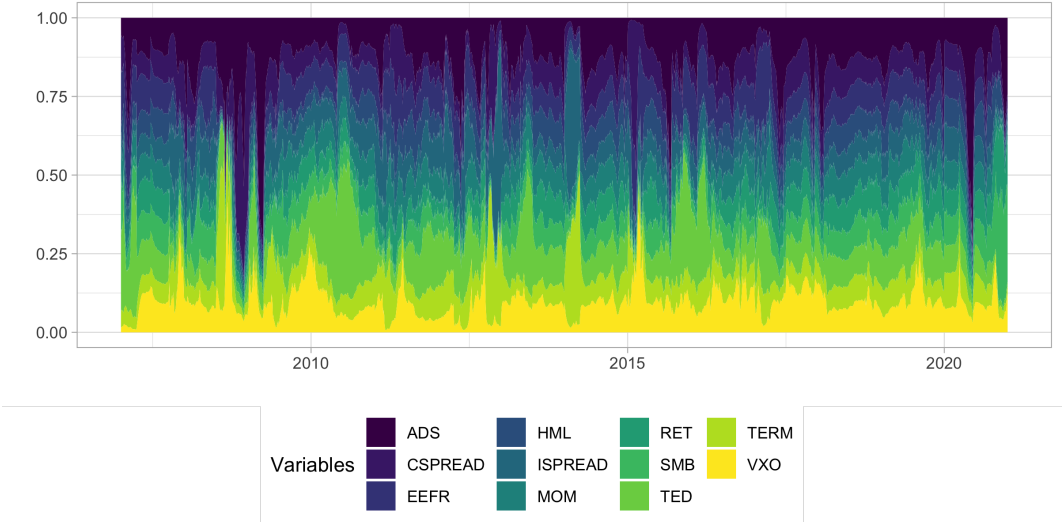

Figure 5: Daily weights for forecast combination.

Group weights for ASGL

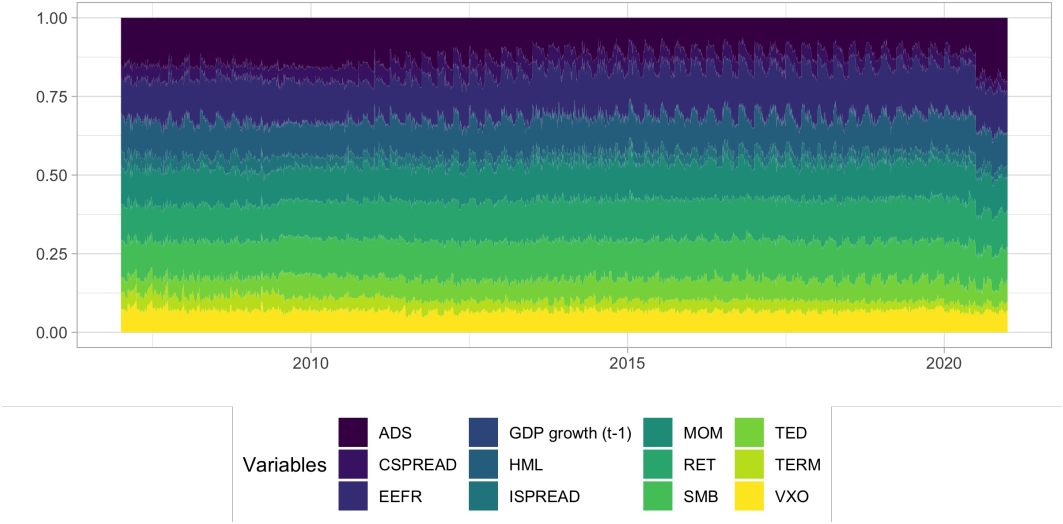

Figure 6: ASGL weights

Evaluation 1: Traditional framework vs our framework

	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	TL	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**
Before COVID-19 (2007Q1 to 2019Q4)										
GaR-MIDAS	0.641	0.001	0.655	0.001	0.653	0.000	0.686	0.001	0.683	0.001
GaR-BMIDAS	0.606	0.000	0.616	0.000	0.631	0.001	**0.643**	**0.001**	**0.654**	**0.001**
GaR-LASSO	**0.59**	**0.001**	**0.559**	**0.000**	**0.569**	**0.000**	0.769	0.145	0.843	0.232
GaR-EN	0.956	0.415	0.978	0.461	0.932	0.366	0.853	0.273	0.858	0.277
GaR-LASSO-PCA	0.617	0.001	0.638	0.002	0.706	0.01	0.83	0.225	0.857	0.266
GaR-EN-PCA	0.617	0.001	0.691	0.01	0.741	0.039	0.809	0.176	0.844	0.251
GaR-ASGL	1.102	0.646	1.037	0.559	0.983	0.471	0.945	0.419	1.221	0.744
Including COVID-19 (2007Q1 to 2020Q4)										
GaR-MIDAS	**0.855**	**0.027**	0.82	0.005	0.804	0.022	0.558	0.094	0.943	0.201
GaR-BMIDAS	0.878	0.021	0.849	0.000	0.839	0.005	0.558	0.087	0.932	0.141
GaR-LASSO	0.864	0.002	0.773	0.006	**0.458**	**0.096**	**0.501**	**0.121**	**0.895**	**0.092**
GaR-EN	0.953	0.243	0.969	0.33	0.822	0.153	0.563	0.139	0.917	0.173
GaR-LASSO-PCA	0.94	0.263	**0.733**	**0.041**	0.488	0.116	0.593	0.133	0.927	0.12
GaR-EN-PCA	0.911	0.102	0.85	0.013	0.841	0.064	0.691	0.116	0.903	0.123
GaR-ASGL	1.106	0.79	1.002	0.506	1.027	0.614	1.056	0.687	1.085	0.766

Figure 7: Relative TL against GaR-CISS

Evaluation 2: Individual vs Combined-GaR for LASSO-Q

	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	TL	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**
Before COVID-19 (2007Q1 to 2019Q4)										
GaR- ISPREAD	2.232	0.978	2.072	0.99	1.768	0.988	1.772	0.997	1.692	0.996
GaR-EEFR	2	0.965	2.392	0.935	1.819	0.972	1.711	0.971	1.649	0.968
GaR-RET	2.107	0.973	2.049	0.962	1.792	0.946	2.2	0.994	1.594	0.971
GaR-SMB	2.079	0.991	2.202	0.991	1.833	0.994	1.93	0.986	2.069	0.99
GaR-HML	2.2	0.957	1.447	0.916	1.57	0.973	1.514	0.968	1.655	0.916
GaR-MOM	**1.448**	**0.987**	2.076	0.959	1.578	0.987	2.072	0.96	1.756	0.973
GaR-VXO	1.813	0.991	1.55	0.995	1.521	0.997	1.258	0.921	1.268	0.886
GaR-CSPREAD	2.242	0.979	1.789	0.993	1.367	1	1.275	0.963	1.259	0.951
GaR-TERM	2.198	0.984	1.946	0.991	1.728	0.989	1.754	0.99	1.684	0.989
GaR-TED	1.643	0.918	1.603	0.968	1.485	0.964	1.44	0.96	1.397	0.931
GaR-ADS	1.678	0.95	1.496	0.91	**1.175**	**0.717**	**0.939**	**0.375**	1.263	0.915
GaR-CISS	1.62	0.999	**1.448**	**0.99**	1.349	0.961	1.236	0.824	**1.185**	**0.749**
Including COVID-19 (2007Q1 to 2020Q4)										
GaR- ISPREAD	1.445	0.968	1.555	0.995	1.53	0.99	1.837	0.973	1.346	0.983
GaR-EEFR	1.449	0.982	1.707	0.992	1.55	0.987	1.855	0.963	1.367	0.979
GaR-RET	1.408	0.992	1.556	0.989	1.503	0.976	1.776	0.964	1.208	0.957
GaR-SMB	1.271	0.948	1.51	0.989	1.301	0.991	1.667	0.96	1.302	0.996
GaR-HML	1.453	0.995	1.504	0.983	1.281	0.913	1.834	0.942	1.3	0.975
GaR-MOM	1.274	0.981	1.712	0.971	1.51	0.969	1.722	0.934	1.307	0.988
GaR-VXO	1.196	0.908	1.335	0.995	1.317	0.994	1.426	0.893	1.129	0.95
GaR-CSPREAD	1.336	0.991	1.351	0.993	1.28	0.939	1.501	0.888	1.133	0.956
GaR-TERM	1.42	0.974	1.502	0.995	1.432	0.993	1.789	0.971	1.334	0.989
GaR-TED	1.315	0.96	1.433	0.994	1.42	0.985	1.731	0.95	1.279	0.98
GaR-ADS	1.375	0.923	**0.595**	**0.174**	**0.655**	**0.159**	**0.504**	**0.152**	**0.743**	**0.26**
GaR-CISS	**1.098**	**0.898**	1.177	0.987	1.188	0.936	1.446	0.884	1.108	0.877

Figure 8: Relative TL against combined-GaR for LASSO-Q

Conclusions

- Our framework can provide an early signal of GDP downturns in pseudo real-time that works well for both, the GFC and the Covid-19 episodes.
- VXO and CSPREAD are especially relevant across models in around the GFC, which highlights the prominent role of uncertainty in determining economic outcomes.
- Financial indicators alone were unable to forecast GDP low quantiles during Covid-19. Indeed, only by including the ADS index we managed to gauge both the sign and the magnitude of the downside GDP risk in this period.

Thank you! Comments welcome at igarron@ub.edu.